

Zhongtian Zhang

Department of Geosciences, Princeton University
Briger Hall, 11 Ivy Lane, Princeton NJ 08544
E-mail: zhongtian.zhang@princeton.edu

Appointments

2025–2027 **Harry Hess Postdoctoral Fellow**, Department of Geosciences, Princeton University
2023–2025 **Carnegie Postdoctoral Fellow**, Earth and Planets Laboratory, Carnegie Science

Education

2023 **Ph.D.**, Earth & Planetary Sciences, Yale University
2019 **M.Phil.**, Geology & Geophysics, Yale University
2017 **B.Sc.**, Geology, Nanjing University

Publications

Articles: Submitted

[19] Grewal, D. S., **Zhang, Z.**, Drażkowska, J. Size-dependent sorting governed planetesimal compositions from the onset of solar system formation. Submitted.

Articles: Published or In-Press

[18] Grewal, D., **Zhang, Z.**, 2026. Beyond the Urey-Craig diagram: A ternary Fe framework for aqueous redox evolution of early solar system planetesimals. *Geochim. Cosmochim. Acta* 277. <https://doi.org/10.1016/j.gca.2026.08.036>.

[17] **Zhang, Z.**, Grewal, D. S., 2026. A re-assessment of moderately volatile elements in the non-matrix component of carbonaceous chondrites. *Icarus*. 452, 117049. <https://doi.org/10.1016/j.icarus.2026.117049>.

[16] Peng, Y., Kislyakova, K., Zheng, D., **Zhang, Z.**, Deng, J., 2025. Induction heating in super-Earths: A thermochemical perspective. *Astrophys. J. Lett.* 933, L56. <https://doi.org/10.3847/2041-8213/ae14e9>.

[15] Grewal, D. S., **Zhang, Z.**, Manilal, V., Kruijer, T., Bottke, W., Stewart, S., 2025. Protracted core formation and impact disruptions shaped the earliest outer solar system planetesimals. *Sci. Adv.* 11 (40), eadw1668. <https://doi.org/10.1126/sciadv.adw1668>.

[14] **Zhang, Z.**, Luo, H., Hao, M., Deng, J., 2025. Regimes of element transfer between Earth's core and basal magma ocean. *J. Geophys. Res., Solid Earth* 130 (7), e2025JB031357. <https://doi.org/10.1029/2025JB031357>

- [13] **Zhang, Z.**, Driscoll, P. E., 2025. Inefficient loss of moderately volatile elements from exposed planetesimal magma oceans. *J. Geophys. Res., Planets* 130 (2), e2024JE008671. <https://doi.org/10.1029/2024JE008671>
- [12] **Zhang, Z.**, Wang, J., 2024. Quantification of classical and non-classical crystallization pathways in calcite precipitation. *Earth Planet. Sci. Lett.* 636, 118712. <https://doi.org/10.1016/j.epsl.2024.118712>.
- [11] **Zhang, Z.**, 2024. Trace elements in IVA iron meteorites explained by limited solid-liquid equilibration during inward solidification. *Icarus*. 408, 115860. <https://doi.org/10.1016/j.icarus.2023.115860>.
- [10] **Zhang, Z.**, 2023. Ice sublimation in planetesimals formed at the outward migrating snowline. *Astrophys. J. Lett.* 956 (1), L25. <https://doi.org/10.3847/2041-8213/acfdaa>
- [9] **Zhang, Z.**, Bercovici, D., 2023. Generation of a measurable magnetic field in a metal asteroid with a rubble-pile inner core. *Proc. Natl. Acad. Sci.* 120 (32), e2221696120. <https://doi.org/10.1073/pnas.2221696120> (Press releases: [Yale News](#), [APS Physics Magazine](#))
- [8] Gong, Z., Evans, D. A. D., **Zhang, Z.**, Yan, C., 2023. Mid-Proterozoic geomagnetic field was more consistent with a dipole than a quadrupole. *Geology* 51 (6), 571–575. <https://doi.org/10.1130/G50941.1>
- [7] **Zhang, Z.**, Bercovici, D., Elkins-Tanton, L. T., 2023. Melt migration in rubble-pile planetesimals: Implications for the formation of primitive achondrites. *Earth Planet. Sci. Lett.* 605, 118019. <https://doi.org/10.1016/j.epsl.2023.118019>
- [6] **Zhang, Z.**, Bercovici, D., Elkins-Tanton, L. T., 2022. Cold compaction and macro-porosity removal in rubble-pile asteroids: 2. Applications. *J. Geophys. Res., Planets* 127 (10), e2022JE007343. <https://doi.org/10.1029/2022JE007343>
- [5] **Zhang, Z.**, Bercovici, D., Elkins-Tanton, L. T., 2022. Cold compaction and macro-porosity removal in rubble-pile asteroids: 1. Theory. *J. Geophys. Res., Planets* 127 (10), e2022JE007342. <https://doi.org/10.1029/2022JE007342>
- [4] **Zhang, Z.**, Bercovici, D., Jordan, J.S., 2021. A two-phase model for the evolution of planetary embryos with implications for the formation of Mars. *J. Geophys. Res., Planets* 126 (4), e2020JE006754. <https://doi.org/10.1029/2020JE006754> (Editor's highlight)
- [3] **Zhang, Z.**, Karato, S.-I., 2021. The effect of pressure on grain-growth kinetics in olivine aggregates with some geophysical applications. *J. Geophys. Res., Solid Earth* 126 (2), e2020JB020886. <https://doi.org/10.1029/2020JB020886>
- [2] **Zhang, Z.**, Wu, B., Wang, T., Hui, H., 2020. Settling of immiscible droplets: A theoretical model for the missing link between microscopic and outcrop observations. *J. Geophys. Res., Solid Earth* 125 (6), e2019JB018829. <https://doi.org/10.1029/2019JB018829>
- [1] Xu, Y., Tang, W., Hui, H., Rudnick, R. L., Shang, S., **Zhang, Z.**, 2019. Reconciling the discrepancy between the dehydration rates in mantle olivine and pyroxene during xenolith emplacement. *Geochim. Cosmochim. Acta* 267, 179–195. <https://doi.org/10.1016/j.gca.2019.09.023>

Invited seminars

- 2026 Department of Earth, Environmental and Planetary Sciences, Brown University
Department of Earth and Planetary Science, University of California, Berkeley
- 2025 Department of Earth, Atmospheric, and Planetary Sciences, Massachusetts Institute of Technology
Department of Earth and Planetary Sciences, Rutgers University
Department of Geophysical Sciences, The University of Chicago
Earth and Planets Laboratory, Carnegie Institution for Science
- 2024 Department of Geosciences, Princeton University
Department of Mineral Sciences, Smithsonian National Museum of Natural History
Department of Geosciences, Stony Brook University
Division of Geological and Planetary Sciences, California Institute of Technology

Honors

- 2025 **Harry H. Hess Postdoctoral Fellowship** (Department of Geosciences, Princeton University)
- 2023 **Carnegie Postdoctoral Fellowship** (Earth and Planets Lab, Carnegie Institution for Science)
- 2023 **Dissertation with Distinction** (Graduate School of Arts and Sciences, Yale University)
- 2022 **Elias Loomis Prize** (Department of Earth & Planetary Sciences, Yale University)

Teaching experience

Teaching Fellowship at Yale University

- Fall 2020 EPS 456/556 Introduction to Seismology
- Spring 2020 G&G275b/F&ES716b Renewable Energy
- Fall 2019 G&G 428/528 Science of Complex Systems

Service

Peer review for Journal of Geophysical Research: Planets (5), Science Advances (2), Nature Astronomy (1), Geophysical Research Letters (2), Geochimica et Cosmochimica Acta (1), Icarus (1), Physics of the Earth and Planetary Interiors (2), Communications Earth and Environments (1)

Proposal panelist for NASA (2)

Proposal reviewer (non-panelist) for NASA (1), European Research Council (1)

Session conveyor for AGU Fall Meeting, DI009: Insights into planetary formation and evolution from interdisciplinary perspectives (2024); Goldschmidt, 01e: Magmatism and surface processes on planetary bodies (2026)

Colloquium committee, Department of Earth & Planetary Sciences, Yale University (2018–2021)

Treasurer, Dana Club, Department of Earth & Planetary Sciences, Yale University (2019–2020)

IC-FG (International Community and First Generation students) working group, IDEA (Inclusion, Diversity, Equity, Anti-racism and Anti-discrimination) committee of Department of Earth & Planetary Sciences, Yale University (2022–2023)

DiMhCi (Disability, Mental health, and Chronic illness) working group, IDEA committee of Department of Earth & Planetary Sciences, Yale University (2022–2023)